

## \* Successive differentiation

$$1. y = \frac{1}{ax+b} \Rightarrow y_n = \frac{(-1)^n a^n n!}{(ax+b)^{n+1}}$$

$$2. y = e^{ax} \Rightarrow y_n = a^n e^{ax}$$

$$3. y = \sin(ax+b) \Rightarrow y_n = a^n \sin\left(ax+b + \frac{n\pi}{2}\right)$$

$$4. y = \cos(ax+b) \Rightarrow y_n = a^n \cos\left(ax+b + \frac{n\pi}{2}\right)$$

$$5. y = e^{ax} \sin(bx+c) \Rightarrow y_n = (a^2+b^2)^{n/2} e^{ax} \sin\left(bx+c + n \tan^{-1} \frac{b}{a}\right)$$

$$6. y = e^{ax} \cos(bx+c) \Rightarrow y_n = (a^2+b^2)^{n/2} e^{ax} \cos\left(bx+c + n \tan^{-1} \frac{b}{a}\right)$$

$$7. y = \log(ax+b) \Rightarrow y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$$

$$8. y = a^{bx} \Rightarrow y_n = b^n (\log a)^n \cdot a^{bx}$$

$$9. y = x^n \Rightarrow y_n = n!$$

## 10. Leibnitz theorem

$$(UV)_n = U_n V + {}^nC_1 U_{n-1} V_1 + {}^nC_2 U_{n-2} V_2 + \dots + {}^nC_n U V_n$$

## 11. Taylor's theorem

$$f(a+h) = f(a) + h f'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{(n-1)}(a) + \dots$$

## 11. Taylor's Theorem

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^n}{n!} f^{(n)}(a+oh)$$

## 12. Maclaurin's Theorem

$$f(a) = f'(a) + af'(a) + \frac{a^2}{2!} f''(a) + \dots + \frac{a^n}{n!} f^{(n)}(ah)$$

## \* Infinite Series

1.  $\lim_{n \rightarrow \infty} n^{1/n} = 1$

2.  $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$

3.  $\lim_{n \rightarrow \infty} \log n = 0$

4.  $\lim_{n \rightarrow 0} \frac{\sin n}{n} = 1$

## \* Tests for convergence &amp; divergence

## 1. Sequence of partial sums (SOPS)

If  $S_n$  convgs, then  $U_n$  also convgs.

## 2. Geometric series

$$\sum ar^{n-1} ; |r| < 1 \rightarrow \text{convg}$$

$$|r| \geq 1 \rightarrow \text{divg.}$$

## 3. Limit Comparison Test (LCT)

$$\lim_{n \rightarrow \infty} \frac{U_n}{V_n} \text{ is finite}$$

if  $V_n \rightarrow \text{convgs}$ , then  $U_n \rightarrow \text{convgs}$

$V_n \rightarrow \text{divgs}$ , then  $U_n \rightarrow \text{divgs}$

## 4. p-series test

$$\sum \frac{1}{n^p} \quad \text{if } p > 1 \rightarrow \text{convg}$$

$$\text{if } p \leq 1 \rightarrow \text{divg.}$$



5. Cauchy  $n^{\text{th}}$  Root test

$$\lim_{n \rightarrow \infty} (U_n)^{1/n} = l ; \quad \begin{array}{l} l < 1 \rightarrow \text{convg} \\ l > 1 \rightarrow \text{divg} \\ l = 1 \rightarrow \text{test fails} \end{array}$$

## 6. Ratio Test

$$\lim_{n \rightarrow \infty} \frac{U_n}{U_{n+1}} = l ; \quad \begin{array}{l} l > 1 \rightarrow \text{convg} \\ l < 1 \rightarrow \text{divg} \\ l = 1 \rightarrow \text{test fails} \end{array}$$

## 7. Raabe's Test

$$\lim_{n \rightarrow \infty} n \left( \frac{U_n}{U_{n+1}} - 1 \right) = l ; \quad \begin{array}{l} l > 1 \rightarrow \text{convg} \\ l < 1 \rightarrow \text{divg} \\ l = 1 \rightarrow \text{test fails} \end{array}$$

## 8. Logarithmic Test

$$\lim_{n \rightarrow \infty} n \left( \frac{\log U_n}{\log U_{n+1}} \right) = l ; \quad \begin{array}{l} l > 1 \rightarrow \text{convg} \\ l < 1 \rightarrow \text{divg} \\ l = 1 \rightarrow \text{fails} \end{array}$$

## 9. Gauss Test

$$\frac{U_n}{U_{n+1}} = 1 + \frac{\lambda}{n} + o\left(\frac{1}{n^2}\right) ; \quad \begin{array}{l} \lambda > 1 \rightarrow \text{convg} \\ \lambda \leq 1 \rightarrow \text{divg} \end{array}$$

## \* Expansions

$$1. \sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots$$

$$2. \cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots$$

$$3. \tan \theta = \theta + \frac{\theta^3}{3!} + \frac{2}{15} \theta^5 + \dots$$

$$4. (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$5. e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$6. \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$7. \log(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \dots\right)$$

### \* Radius of curvature

#### \* Absolute convergence

~~to be done~~

Let a series be  $U_1 + U_2 + U_3 + \dots + U_n$

→ If  $|U_1| + |U_2| + |U_3| + \dots + |U_n|$  is convg then  $\sum U_n$  is absolutely convgt.

→ If  $\sum |U_n|$  is divg. but  $\sum U_n$  is convg, then  $\sum U_n$  is conditionally convgt.

### \* Radius of curvature

1. Cartesian curve  $y = f(x)$

$$\rho = \frac{(1 + y_1'^2)^{3/2}}{y_2'}$$

2. Parametric curve  $y = f(t)$   $x = g(t)$

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''}$$

3. Polar curve  $r = f(\theta)$

$$\rho = \frac{(r_1^2 + r_2^2)^{3/2}}{r_1^2 + 2r_1r_2 - r_2^2}$$



## 4. Newton's method

[when  $x$ -axis is tangent]

$$\rho = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2}{2y}$$

$$\rho = \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{y^2}{2x}$$

[when  $y$ -axis is tangent]5. Expansion method  $y = f(x)$ [ $y = x$  is tangent]

$$\text{Let } y = px + \frac{q}{2} x^2 + \dots$$

$$\text{where } p = f'(0) \quad q = f''(0)$$

$$\therefore \rho = \frac{(1 + p^2)^{3/2}}{q}$$

## \* Area under curve

$$1. \quad y = f(x), \quad x = a, \quad x = b$$

$$A = \int_a^b y \, dx$$

$$2. \quad x = f(y), \quad y = a, \quad y = b$$

$$A = \int_a^b x \, dy$$

$$3. \quad x = \phi(t), \quad y = f(t), \quad x\text{-axis}$$

$$A = \int_{t_1}^{t_2} y \cdot \frac{dx}{dt} \cdot dt$$

$$4. \quad x = \phi(t), \quad y = f(t), \quad y\text{-axis}$$

$$A = \int_{t_1}^{t_2} x \cdot \frac{dy}{dt} \cdot dt$$

5.  $r = f(\theta)$

$$A = \frac{1}{2} \int_a^b r^2 d\theta$$

\* Length of curve

1.  $y = f(x) \quad x = a \quad x = b$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

2.  $x = f(y) \quad y = a \quad y = b$

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

3.  $x = f(t) \quad y = g(t)$

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

4.  $r = f(\theta)$

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

\* Volume of revolution

1. Revolution about  $x$ -axis  $\Rightarrow \int_a^b \pi y^2 dx$

2. Revolution about  $y$ -axis  $\Rightarrow \int_a^b \pi x^2 dy$



3. Revolution about initial line ( $\theta=0$ )  $\Rightarrow \int_a^b \frac{2\pi}{3} r^3 \sin \theta d\theta$

4. Revolution about ( $\theta = \frac{\pi}{2}$ )  $\Rightarrow \int_a^b \frac{2\pi}{3} r^3 \cos \theta d\theta$

\* Surface area of Revolution

1. Revolution about x-axis  $\Rightarrow \int_a^b 2\pi y \frac{dl}{dn} \cdot dn$

2. Revolution about y-axis  $\Rightarrow \int_a^b 2\pi x \frac{dl}{dy} \cdot dy$

\* Differential Eq<sup>n</sup>s.

1. Linear eq<sup>n</sup>s.

$$\frac{dy}{dx} + P_y = Q$$

$$\Rightarrow IF = e^{\int P_y dx}$$

$$\text{Sol<sup>n</sup> } y(IF) = \int Q \times IF dx$$

2. Exact diff. eq<sup>n</sup>  $\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad (Mdx + Ndy = 0)$

$$\text{Sol<sup>n</sup> } \int_{y \text{ const}} M dx + \int N^* dy = C$$

If  $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ , then we will reduce it:

$\rightarrow$  Homo. eq<sup>n</sup>  $\Rightarrow IF = \frac{1}{Mx + Ny}$

$$\rightarrow P_1(x,y)ydx + P_2(x,y)x dy = 0 ; IF = \frac{1}{Mx - Ny}$$

$$\rightarrow \text{for } Mdx + Ndy = 0$$

$$\text{if } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x), \text{ then } e^{\int f(x) dx} \text{ is IF}$$

$$\text{if } \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y), \text{ then } e^{\int f(y) dy} \text{ is IF}$$

### \* Method of Variation Parameters

$\rightarrow$  For find particular integral (PI)

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$PI = -y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx$$



### 13.7 WORKING PROCEDURE TO SOLVE THE EQUATION

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + k_{n-1} \frac{dy}{dx} + k_n y = X$$

of which the symbolic form is

$$(D^n + k_1 D^{n-1} + \dots + k_{n-1} D + k_n) y = X.$$

#### Step I. To find the complementary function

(i) Write the A.E.

$$D^n + k_1 D^{n-1} + \dots + k_{n-1} D + k_n = 0 \text{ and solve it for } D.$$

i.e.,

(ii) Write the C.F. as follows :

#### Roots of A.E.

1.  $m_1, m_2, m_3 \dots$  (real and different roots)
2.  $m_1, m_1, m_3 \dots$  (two real and equal roots)
3.  $m_1, m_1, m_1, m_4 \dots$  (three real and equal roots)
4.  $\alpha + i\beta, \alpha - i\beta, m_3 \dots$  (a pair of imaginary roots)
5.  $\alpha \pm i\beta, \alpha \pm i\beta, m_5 \dots$  (2 pairs of equal imaginary roots)

#### C.F.

$$c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \dots$$

$$(c_1 + c_2 x) e^{m_1 x} + c_3 e^{m_3 x} + \dots$$

$$(c_1 + c_2 x + c_3 x^2) e^{m_1 x} + c_4 e^{m_4 x} + \dots$$

$$e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) + c_3 e^{m_3 x} + \dots$$

$$e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x] + c_5 e^{m_5 x} + \dots$$